

Homework 8

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Summary

A mathematical model for displacement of 3-UPU parallel machine is set up and solved. Local displacement at each joint is solved. The model is verified.

Problem Set up

Simplify the schematic of parallel machine described in HW6 (see Figure 1), re-orientate it and change the coordinate frame set up into Figure 2.



Figure 1 Parallel machine described in HW6. It is designed to be actuated by linear actuator.

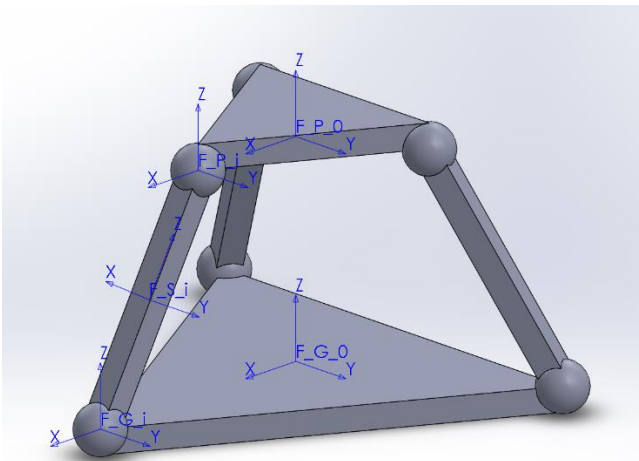


Figure 2 Simplified schematic of 3-UPU parallel machine. The bag-like part indicates the compliant universal joint. The displacement of three linear actuator happen to be the same in this figure

Name three UPU chains of parallel machine with index $i \in \{1,2,3\}$. For each chain $i \in \{1,2,3\}$, $\mathcal{G}_i, \mathcal{S}_i, \mathcal{P}_i$ indicates the local body frame of ground universal joint, slider, and platform universal joint.

$\mathcal{G}_0$ indicates the global ground frame. $\mathcal{P}_0$ indicates the global platform frame.

Equation Setting Up

(a) We want to express local angular displacement at each joint with characteristic length (eq. 1).

$$\begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=1} \quad \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=2} \quad \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=3}$$

$$= h(L_{\mathcal{S}_1} \quad L_{\mathcal{S}_3} \quad L_{\mathcal{S}_3} \quad L_{\mathcal{P}} \quad L_{\mathcal{G}}) \quad (\text{eq. 1})$$

The definition of terms on chain i are as follows (Figure 3).

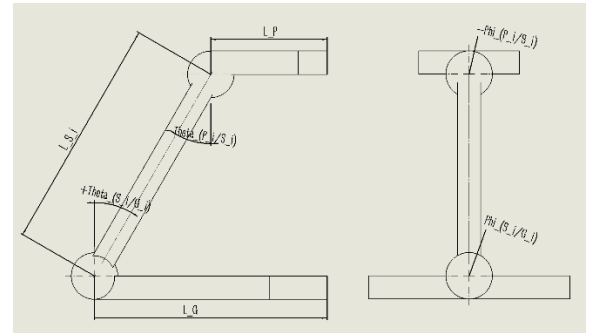


Figure 3 In this figure, $\Phi_{\mathcal{S}_i/\mathcal{G}_i}$ and $\Phi_{\mathcal{P}_i/\mathcal{S}_i}$ happens to be 0, but generally they are nonzero.

Then, let's derive and solve equations of orientation and displacement. Together, they will form the complete and expanded equation in (eq. 1)

(b) Equation of Orientation:

Local angular displacement at joint is described by pitch Θ and roll Φ angles. We have 6 universal joint and thus got 12 unknown displacement. Orientation relation provides 6 equations.

Universal joints constraint yaw Ψ but allow pitch Θ and roll Φ . Thus, the sequence of frame $\{\mathcal{G}_0, \mathcal{G}_i, \mathcal{S}_i, \mathcal{P}_i, \mathcal{P}_0\}$ is defined in such a way that the only yaw Ψ happens between global frame and local frame, and there is only pitch and roll Φ, Θ between local frames.

As a result, the directional cosine $\mathcal{F}$ of frame sequence has relation in (eq. 2)

$$\mathcal{F}_{\mathcal{G}_0} \xrightarrow{\mathcal{O}(\Psi_{\mathcal{G}_i/\mathcal{G}_0}, 0, 0)} \mathcal{F}_{\mathcal{G}_i} \xrightarrow{\mathcal{O}(0, \Theta_{\mathcal{S}_i/\mathcal{G}_i}, \Phi_{\mathcal{S}_i/\mathcal{G}_i})} \mathcal{F}_{\mathcal{S}_i}$$

$$\xrightarrow{\mathcal{O}(0, \Theta_{\mathcal{P}_i/\mathcal{S}_i}, \Phi_{\mathcal{P}_i/\mathcal{S}_i})} \mathcal{F}_{\mathcal{P}_i} \xrightarrow{\mathcal{O}(\Psi_{\mathcal{P}_0/\mathcal{P}_i}, 0, 0)} \mathcal{F}_{\mathcal{P}_0} \quad (\text{eq. 2})$$

Since the ground and platform are rotational symmetry about their own axis, we thus can set the yaw angle Ψ as constant in (eq. 3)

$$\Psi_{\mathcal{G}_i/\mathcal{G}_0} = -\Psi_{\mathcal{P}_0/\mathcal{P}_i} = \frac{2i\pi}{3}, \quad i \in \{1, 2, 3\} \quad (\text{eq. 3})$$

As a result, the final orientation matrix $\mathcal{O}_{\mathcal{P}_0/\mathcal{G}_0}$ describing the rotation from ground frame $\mathcal{F}_{\mathcal{G}_0}$ to the platform frame $\mathcal{F}_{\mathcal{P}_0}$ is a function (name it $f: \mathbb{R}^4 \rightarrow \mathbb{R}^{3 \times 3}$) about and only about $\Theta_{\mathcal{S}_i/\mathcal{G}_i}, \Phi_{\mathcal{S}_i/\mathcal{G}_i}, \Theta_{\mathcal{P}_i/\mathcal{S}_i}, \Phi_{\mathcal{P}_i/\mathcal{S}_i}$ in (eq. 4)

$$\mathcal{O}_{\mathcal{P}_0/\mathcal{G}_0} = \mathcal{O}\left(\frac{2i\pi}{3}, 0, 0\right) f\left(\begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}\right) \mathcal{O}\left(-\frac{2i\pi}{3}, 0, 0\right) \quad (\text{eq. 4})$$

In above derivation, we focus on i^{th} chain (Figure 4).

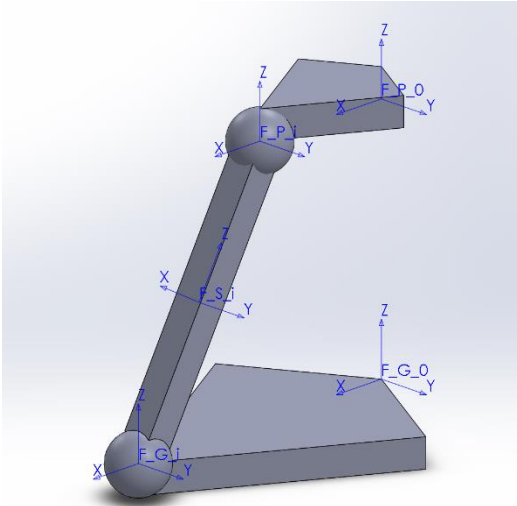


Figure 4 We focus on i^{th} chain

However, all 3 chains should have same final orientation matrix $\mathcal{O}_{\mathcal{P}_0/\mathcal{G}_0}$. Thus, (eq. 4) provides us with 2 matrix equations shown in (eq. 5):

$$\begin{aligned} & \mathcal{O}\left(\frac{2i\pi}{3}, 0, 0\right) f\left(\begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}\right) \mathcal{O}\left(-\frac{2i\pi}{3}, 0, 0\right) \Bigg|_{i=1} \\ &= \mathcal{O}\left(\frac{2i\pi}{3}, 0, 0\right) f\left(\begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}\right) \mathcal{O}\left(-\frac{2i\pi}{3}, 0, 0\right) \Bigg|_{i=2} \end{aligned}$$

$$= \mathcal{O}\left(\frac{2i\pi}{3}, 0, 0\right) f\left(\begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}\right) \mathcal{O}\left(-\frac{2i\pi}{3}, 0, 0\right) \Bigg|_{i=3} \quad (\text{eq. 5})$$

(c) Equation of Displacement:

Resolve the displacement vector from center point of ground G_0 to center point of platform P_0 into ground frame $\mathcal{G}_0$. The notation of resolved vector is $G_0 P_0|_{\mathcal{G}_0}$, which can be expanded as equation (eq. 6)

$$\begin{aligned} G_0 P_0|_{\mathcal{G}_0} = & \mathcal{O}_{\mathcal{G}_i/\mathcal{G}_0}^{-1} \left[\begin{pmatrix} L_G \\ 0 \\ 0 \end{pmatrix}_{\mathcal{G}_i} + \mathcal{O}_{\mathcal{S}_i/\mathcal{G}_i}^{-1} \left[\begin{pmatrix} 0 \\ 0 \\ L_{S_i} \end{pmatrix}_{\mathcal{S}_i} + \mathcal{O}_{\mathcal{P}_i/\mathcal{S}_i}^{-1} \begin{pmatrix} -L_P \\ 0 \\ 0 \end{pmatrix}_{\mathcal{P}_i} \right] \right] \quad (\text{eq. 6}) \end{aligned}$$

Where we have (eq. 7)

$$\begin{bmatrix} \mathcal{O}_{\mathcal{G}_i/\mathcal{G}_0}^{-1} \\ \mathcal{O}_{\mathcal{S}_i/\mathcal{G}_i}^{-1} \\ \mathcal{O}_{\mathcal{P}_i/\mathcal{S}_i}^{-1} \end{bmatrix} = \begin{bmatrix} \mathcal{O}(2i\pi/3, 0, 0)^T \\ \mathcal{O}(0, \Theta_{\mathcal{S}_i/\mathcal{G}_i}, \Phi_{\mathcal{S}_i/\mathcal{G}_i})^T \\ \mathcal{O}(0, \Theta_{\mathcal{P}_i/\mathcal{S}_i}, \Phi_{\mathcal{P}_i/\mathcal{S}_i})^T \end{bmatrix} \quad (\text{eq. 7})$$

Similar to equation of orientation, all 3 chains should have same final displacement. We thus get 2 matrix equation from (6) for $i \in \{1, 2, 3\}$

Numerical Solution

Above derivation gives 12 variables and 24 scalar equations (eq. 5), (eq. 6). There are more equations than variables. In this case, it's more realistic to solve for a least-squared fitted numerical solution than to go for an analytic solution.

Result

The solver managed to solve for local angular displacement at joint (eq. 8) with given input (eq. 9).

$$\begin{bmatrix} \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=1} & \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=2} & \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{G}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{G}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=3} \end{bmatrix} \quad (\text{eq. 8})$$

$$(L_{S_1} \quad L_{S_3} \quad L_{S_3} \quad L_P \quad L_G) \quad (\text{eq. 9})$$

With a fixed input value, only 3 set of displacement solution is found. For example, given input (eq. 10), only three solution is found (Figure 5).

$$(L_{S_1} \quad L_{S_3} \quad L_{S_3} \quad L_P \quad L_G) = (1 \quad 1 \quad 0.9 \quad 0.5 \quad 1) \quad (\text{eq. 10})$$

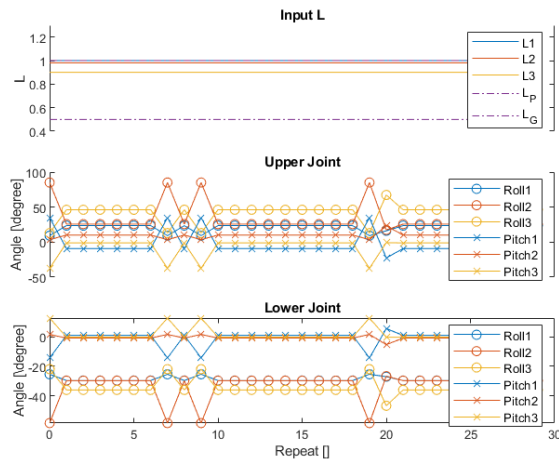


Figure 5 For one same input, the numerical solver is run for 30 times with rand initial guess. Only three displacement solution is found.

Given input in (eq.10) , only 3 displacement solutions are found. This means that our 3-UPU parallel machine is fully constraint by actuators. That's because, any under-constraint mechanism will infinite many displacement solutions, while a fully-constraint mechanism can have more than 1 displacement solution (Figure 6).

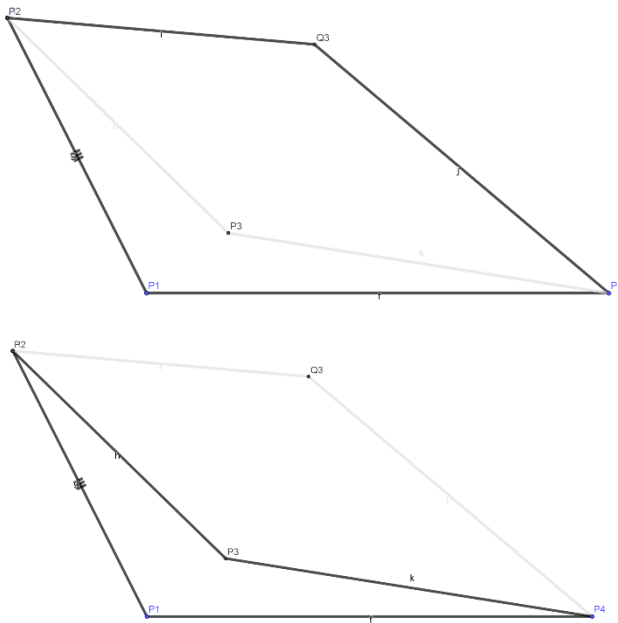


Figure 6 Analogy of 4 bar linkage. Given an input angular displacement, there are at most two set of possible displacement solution. However, as long as the mechanism does not reach its toggle position, it will always in one set of displacement solution. As a result, though the displacement solution is not unique, the mechanism is still fully-constraint

The visualization of 3 displacement solutions are as follows (Figure 7, 8, 9).

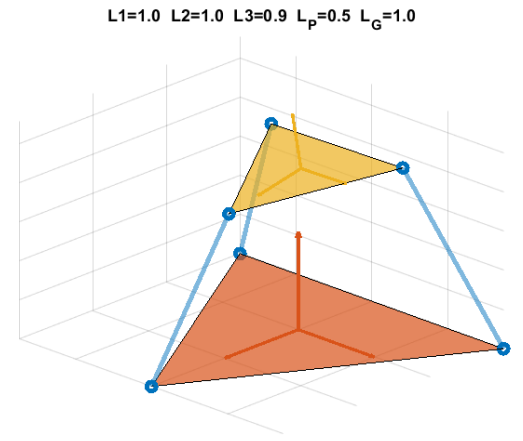


Figure 7 Displacement solution 1. 6 blue circle indicates our universe joint in Figure 1.

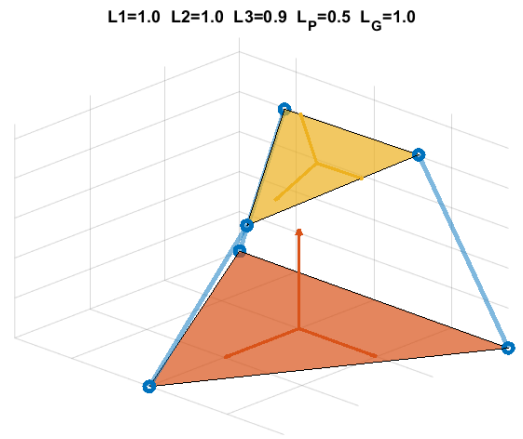


Figure 8 Displacement solution 2

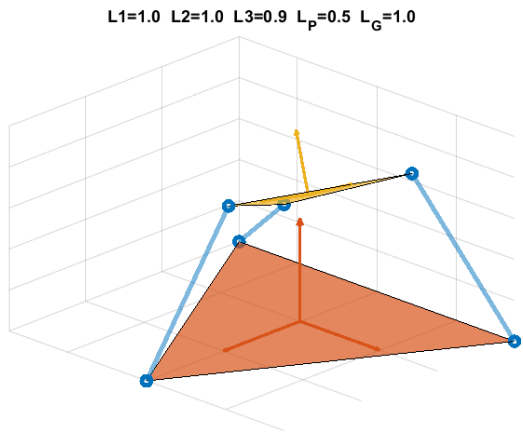


Figure 9 Displacement solution 3

Question

Prof. Kota, could it be that a constraining leg is necessary for a 3-UCU parallel machine instead of a 3-UPU one? I look into Prof. Moon's design again (Figure 10, p.4), and I think maybe his actuators are 3 2-DOF cylinder sliders (C) instead of 3 1-DOF block sliders (P).

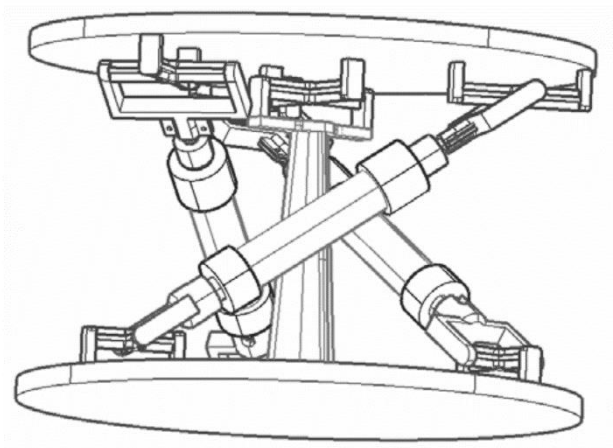


Figure 10 Moon, Y.-M., and Kota, S. Synthesis of compliant parallel kinematic machines

Future Work

Use the solver for design purpose: determine ideal travel of actuator, relative size of platform, initial angular displacement of compliant joint.

Notation

$i \in \{1,2,3\}$	Index of chain
G_0	Position of center of ground
P_0	Position of center of platform
G_i	Position of center point of universal joint fixed on ground in i^{th} chain
P_i	Position of center point of universal joint fixed on platform in i^{th} chain
S_i	Middle point on slider in i^{th} chain
AB	Displacement vector from point A to point B
$\mathcal{X}$	Body frame $\mathcal{X}$
$AB _x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}_x$	Vector AB resolved in frame $\mathcal{X}$
$\mathcal{F}_x$	Directional cosine matrix of body frame whose center point is $\mathcal{X}$
$\mathcal{O}_{x/y} = \mathcal{O}(\Psi_{x/y}, \Theta_{x/y}, \Phi_{x/y})$	Eulerian orientation matrix transforming directional cosine matrix $\mathcal{F}_y$ to $\mathcal{F}_x$ and vector resolved in frame $\mathcal{X}$ to resolved in frame $\mathcal{Y}$ ($\mathcal{F}_x = \mathcal{O}_{x/y}\mathcal{F}_y$, $AB _x = \mathcal{O}_{x/y}AB _y$). The Eulerian angle is in 321 convention.
L_G, L_P, L_S	Link length of ground, platform and slider
$\Phi_{x/y}$	Roll angle from frame $\mathcal{Y}$ to frame $\mathcal{X}$ (Eulerian angle about x axis in frame $\mathcal{Y}$)
$\Theta_{x/y}$	Pitch angle from frame $\mathcal{Y}$ to frame $\mathcal{X}$ (Eulerian angle about y axis in frame $\mathcal{Y}$)
$\Psi_{x/y}$	Yaw angle from frame $\mathcal{Y}$ to frame $\mathcal{X}$ (Eulerian angle about z axis in frame $\mathcal{Y}$)

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$$\begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{E}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{E}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=1} \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{E}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{E}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=2} \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{E}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{E}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=3} = h(L_{S_1} \ L_{S_3} \ L_{S_3} \ L_p \ L_G)$$

Main

```
clear;clc;close all;
tolerance=1e-20;
L_P=0.5;
L_G=1;
Produce L and G_data
N=30;
L_data=[...
    ones(1,1,N)*1,    ones(1,1,N)*0.98,
    reshape(linspace(0.9,0.9,N),1,1,N),
    ones(1,1,N)*L_P,    ones(1,1,N)*L_G];
G_data=[];
for i=1:N
    G_data=cat(3,G_data,L_G*[...
        -1/2        -1/2        1;
        sqrt(3)/2   -sqrt(3)/2  0;
        0            0            0;
    ]);
end
Solver
[x_sol_data,resnorm_data,N]=L2x(L_data,N,tolerance);
[P_data,P0_data,O_PG_data,std_data]=xL2PP0
OPGstd(x_sol_data,L_data,N);
[temp1,temp2,temp3]=dcm2angle(O_PG_data);
Angle_PG_data=[...
    reshape(temp1,1,1,N),...
    reshape(temp2,1,1,N),...
    reshape(temp3,1,1,N)...
];
Plotting
```

```
PMfigure_handle=figure("Name","Parallel
Machine");
n=21;
PMfigure_handle=PMplot(P_data(:,:,n),P0_data(:,:,n),O_PG_data(:,:,n),P
Mfigure_handle,x_sol_data(:,:,n),L_data(:,:,n));
```

```
Statefigure_handle=figure("Name","12
State");
Statefigure_handle=AnglePlot12(x_sol_data,
L_data,Statefigure_handle);
```

Function

Eulerian Angle to Directional Cosin (Allow Syms)

```
function O=myEulZYX(Psi,Theta,Phi)
% Return the double/symbolic orientation
matrix of Eulerian angle yaw,
% pitch and roll
O3=[cos(Psi)    sin(Psi)    0
    -sin(Psi)   cos(Psi)    0
    0           0          1];
O2=[cos(Theta)  0          -sin(Theta)
    0           1           0
    sin(Theta)  0          cos(Theta)];
O1=[1          0          0
    0          cos(Phi)   sin(Phi)
    0          -sin(Phi)  cos(Phi)];
O=O1*O2*O3;
end
```

$$\begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{E}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{E}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=1} \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{E}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{E}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=2} \begin{bmatrix} \Theta_{\mathcal{S}_i/\mathcal{E}_i} \\ \Phi_{\mathcal{S}_i/\mathcal{E}_i} \\ \Theta_{\mathcal{P}_i/\mathcal{S}_i} \\ \Phi_{\mathcal{P}_i/\mathcal{S}_i} \end{bmatrix}_{i=3} = h(L_{S_1} \ L_{S_3} \ L_{S_3} \ L_p \ L_G)$$

24 Equation Handle Generator

```
% function f=overLoaded24Equation(x,L)
% % O_PG=sym(zeros(3,3,3));
% % G0P0=sym(zeros(3,1,3));
% O_PG=[];
% G0P0=[];
% for i=1:3
%     % Equation of Orientation
%     O_PG(:,:,i)=...
%     O_PG=cat(3,O_PG,...
%     myEulZYX(-
% 2*mod(i,3)*pi/3,0,0)*...
%     myEulZYX(0,x(3,i),x(4,i))*...
%     myEulZYX(0,x(1,i),x(2,i))*...
%     myEulZYX(2*mod(i,3)*pi/3,0,0)...
% );
%     % Equation of Displacement
%     G0P0(:,:,i)=...
%     G0P0=cat(3,G0P0,...
%     myEulZYX(2*mod(i,3)*pi/3,0,0)'*[L(5);0;0]+
% ...
%     myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(x(
% 1,i),x(2,i),0)*[0;0;L(i)]+...
```

```
%
myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(x(
1,i),x(2,i),0)'*myEulZYX(x(3,i),x(4,i),0)'
*[-L(4);0;0]...
%
);
% end
% f=[...
%     O_PG(:,:,3)-O_PG(:,:,1);...
%     O_PG(:,:,3)-O_PG(:,:,2);...
%     G0P0(:,:,3)-G0P0(:,:,1)';...
%     G0P0(:,:,3)-G0P0(:,:,2)';...
% ];
% end
```

```
function f=overLoaded24Equation(x,L)
% O_PG=sym(zeros(3,3,3));
% G0P0=sym(zeros(3,1,3));
O_PG=[];
G0P0=[];
for i=1:3
    % Equation of Orientation
%     O_PG(:,:,i)=...
    O_PG=cat(3,O_PG,...
        myEulZYX(-2*mod(i,3)*pi/3,0,0)*...
        myEulZYX(0,x(3,i),x(4,i))*...
        myEulZYX(0,x(1,i),x(2,i))*...
        myEulZYX(2*mod(i,3)*pi/3,0,0)...
    );
    % Equation of Displacement
%     G0P0(:,:,i)=...
    G0P0=cat(3,G0P0,...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*[L(5);0;0]+
...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(0,
x(1,i),x(2,i))'*[0;0;L(i)]+...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(0,
x(1,i),x(2,i))'*myEulZYX(0,x(3,i),x(4,i))'
*[-L(4);0;0]...
);
end
f=[...
    O_PG(:,:,3)-O_PG(:,:,1);...
    O_PG(:,:,3)-O_PG(:,:,2);...
    G0P0(:,:,3)-G0P0(:,:,1)';...
    G0P0(:,:,3)-G0P0(:,:,2)';...
];
end
```

Make x into More Detailed Result

```
% function
[P_data,P0_data,O_PG_data,std_data]=xL2PP0
OPGstd(x_data,L,N)
% P_data=[];
```

```
% P0_data=[];
% O_PG_data=[];
% P0_std=[];
% O_PG_std=[];
% for j=1:N
%     P_single=[];
%     P0_single=[];
%     O_PG_single=[];
%     for i=1:3
%         % Equation of Displacement
%         P_single=cat(2,P_single,...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*[L(:,5,j);0
;0]+...
%
myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(x_
data(1,i,j),x_data(2,i,j),0)'*[0;0;L(:,i,j
)]...
%
);
%     P0_single=cat(2,P0_single,...
%
myEulZYX(2*mod(i,3)*pi/3,0,0)'*[L(:,5,j);0
;0]+...
%
myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(x_
data(1,i,j),x_data(2,i,j),0)'*[0;0;L(:,i,j
)]+...
%
myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(x_
data(1,i,j),x_data(2,i,j),0)'*myEulZYX(x_d
ata(3,i,j),x_data(4,i,j),0)'*[-
L(:,4,j);0;0]...
%
);
%
O_PG_single=cat(3,O_PG_single,...
%     myEulZYX(-
2*mod(i,3)*pi/3,0,0)*...
%
myEulZYX(0,x_data(3,i,j),x_data(4,i,j))*..
.
%
myEulZYX(0,x_data(1,i,j),x_data(2,i,j))*..
.
%
myEulZYX(2*mod(i,3)*pi/3,0,0)...
%
);
%     end
%     P_data=cat(3,P_data,P_single);
%
P0_data=cat(3,P0_data,mean(P0_single,2));
%
O_PG_data=cat(3,O_PG_data,mean(O_PG_single
,3));
%     P0_std=[P0_std;...
%         mean(std(P0_single,0,2))...
%     ];
%     O_PG_std=[O_PG_std;...
%
mean(mean(std(O_PG_single,0,3)))...
%
];
```

```

% end
%
std_data=struct("P0",P0_std,"O_PG",O_PG_std);
% end

function
[P_data,P0_data,O_PG_data,std_data]=xL2PP0
OPGstd(x_data,L,N)
P_data=[];
P0_data=[];
O_PG_data=[];
P0_std=[];
O_PG_std=[];
for j=1:N
    P_single=[];
    P0_single=[];
    O_PG_single=[];
    for i=1:3
        % Equation of Displacement
        P_single=cat(2,P_single,...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*[L(:,5,j);0;0]+...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(0,
x_data(1,i,j),x_data(2,i,j))'*[0;0;L(:,i,j)
])...

        );
        P0_single=cat(2,P0_single,...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*[L(:,5,j);0;0]+...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(0,
x_data(1,i,j),x_data(2,i,j))'*[0;0;L(:,i,j)
])+...

myEulZYX(2*mod(i,3)*pi/3,0,0)'*myEulZYX(0,
x_data(1,i,j),x_data(2,i,j))'*myEulZYX(0,x
_data(3,i,j),x_data(4,i,j))'*[-
L(:,4,j);0;0]...

        );
        O_PG_single=cat(3,O_PG_single,...
        myEulZYX(-
2*mod(i,3)*pi/3,0,0)*...

myEulZYX(0,x_data(3,i,j),x_data(4,i,j))*..
.

myEulZYX(0,x_data(1,i,j),x_data(2,i,j))*..
.

myEulZYX(2*mod(i,3)*pi/3,0,0)...
        );
    end
    P_data=cat(3,P_data,P_single);

P0_data=cat(3,P0_data,mean(P0_single,2));

```

```

O_PG_data=cat(3,O_PG_data,mean(O_PG_single
,3));
    P0_std=[P0_std;...
            mean(std(P0_single,0,2))...
            ];
    O_PG_std=[O_PG_std;...

mean(mean(std(O_PG_single,0,3)))...
        ];
end
std_data=struct("P0",P0_std,"O_PG",O_PG_std);
end

```

Function Main Solver: From L to x

```

function
[x_sol_data,resnorm_data,N]=L2x(L,N,tolera
nce)
y_data=zeros(8,3);
lb=-ones(4,3)*pi/2;
ub=ones(4,3)*pi/2;
x_sol_data=[];
resnorm_data=[];
for i=1:N

option=optimoptions("lsqcurvefit","MaxFunc
tionEvaluations",2e3,"FunctionTolerance",t
olerance,"StepTolerance",tolerance,"Optima
lityTolerance",tolerance);
    x0=rand(4,3);

[x_sol,resnorm]=lsqcurvefit(@overLoaded24E
quation,x0,L(:, :, i),y_data,lb,ub,option);
    x_sol_data=cat(3,x_sol_data,x_sol);

resnorm_data=cat(3,resnorm_data,resnorm);
end
end

```

Parallel Machine Plot

```

function
figure_handle=PMplot(P,P0,O_PG,G,figure_ha
ndle,x,L)

Plot Setting
color_order=[1,2,3];
scale_quiver3=[0.5,0.3];
alpha=[0.7,0.7,0.5];

Plotting
set(0,'CurrentFigure',figure_handle);
color_map=get(gca,'colororder');

Line
for i=1:3
    plot3(...
        [G(1,i),P(1,i)],...
        [G(2,i),P(2,i)],...

```

```

        [G(3,i),P(3,i)],...
        'o-
', 'LineWidth', 3, "Color", [color_map(color_o
rder(1),:),alpha(3)]...
    );
    hold on;
end

Ground
Fill
set(...

fill3(G(1,:),G(2,:),G(3,:),color_map(color
_order(2),:)),...
    'FaceAlpha',alpha(1)...
    );
hold on;

GO Frame
G0_frame_handle=frameplot(eye(3),[0;0;0],f
igure_handle,color_map(color_order(2),:),s
cale_quiver3(1));

% % Interm Frame
% ii=1;
% Gii_frame_handle=frameplot(...
%     myEulZYX(pi*2/3*mod(ii,3),0,0)',...
%     G(:,ii),...
%
figure_handle,color_map(color_order(2),:),
scale_quiver3(1));
% Sii_frame_handle=frameplot(...
%     (...
%     myEulZYX(0,x(1,ii),x(2,ii))*...
%
myEulZYX(pi*2/3*mod(ii,3),0,0)...
%     )',...
%     0.5*G(:,ii)+0.5*P(:,ii),...
%
figure_handle,color_map(color_order(2),:),
scale_quiver3(1));
% Pii_frame_handle=frameplot(...
%     (...
%     myEulZYX(0,x(3,ii),x(4,ii))*...
%     myEulZYX(0,x(1,ii),x(2,ii))*...
%
myEulZYX(pi*2/3*mod(ii,3),0,0)...
%     )',...
%     P(:,ii),...
%
figure_handle,color_map(color_order(2),:),
scale_quiver3(1));
% %

Platform
Fill
set(...

fill3(P(1,:),P(2,:),P(3,:),color_map(color
_order(3),:)), ...
    'FaceAlpha',alpha(2)...

```

```

    );
    hold on;

```

```

Frame
ii=1;
P0_frame_handle=frameplot(...
    (...
        myEulZYX(-
pi*2/3*mod(ii,3),0,0)*...
        myEulZYX(0,x(3,ii),x(4,ii))*...
        myEulZYX(0,x(1,ii),x(2,ii))*...
        myEulZYX(pi*2/3*mod(ii,3),0,0)...
    )',...
    P0,...

figure_handle,color_map(color_order(3),:),
scale_quiver3(2));

% Axis and Grid
set(gca,'XColor','none','YColor','none','Z
Color','none');
axis equal;
grid on;
view(134,22);
set(figure_handle,'Visible','On');
title(sprintf("L1=%.1f L2=%.1f L3=%.1f
L_P=%.1f
L_G=%.1f",L(1),L(2),L(3),L(4),L(5)))
end

```

Function: Frame Plot

```

function
frame_handle=frameplot(dcm,origin,figure_h
andle,color_map_single,scale_quiver3_singl
e)
set(0,'CurrentFigure',figure_handle);
temp_0_quiver3=[origin,origin,origin];
frame_handle=quiver3(...

temp_0_quiver3(1,:),temp_0_quiver3(2,:),te
mp_0_quiver3(3,:),...
    dcm(1,:),dcm(2,:),dcm(3,:),...
    scale_quiver3_single,...

'Color',color_map_single,'lineWidth',2);
hold on;
end

```

Function: Angle Plot 12

```

function
figure_handle=AnglePlot12(x_data,L_data,fi
gure_handle)
set(0,'CurrentFigure',figure_handle);
color_map=get(gca,'ColorOrder');
N=size(L_data,3);
temp_valuex=[1:1:N]-1;

```

```

% Input L_data
subplot(3,1,1);
for i=1:3

plot(temp_valuex,reshape(L_data(1,i,:),N,1),
'-','color',color_map(i,:));
    hold on;
end
for i=4:5

plot(temp_valuex,reshape(L_data(1,i,:),N,1),
'-.','color',color_map(4,:));
    hold on;
end
ylabel("L");
ylim([0.4 1.3]);
legend(["L1";"L2";"L3";"L_P";"L_G"]);
title("Input L");
set(gca,'XColor','none');


% Upper Joint
subplot(3,1,2);
for i=1:3

plot(temp_valuex,180/pi*reshape(x_data(3,i,:),N,1),'o-','Color',color_map(i,:));
    hold on;
end
for i=1:3
    plot(temp_valuex,-
180/pi*reshape(x_data(4,i,:),N,1),'x-','Color',color_map(i,:));
    hold on;
end
legend([...
    "Roll"+num2str([1:1:3]');...
    "Pitch"+num2str([1:1:3]')...
    ]);
ylabel("Angle [\degree]");
title("Upper Joint");
set(gca,'XColor','none');


% Lower Joint
subplot(3,1,3);
for i=1:3

plot(temp_valuex,180/pi*reshape(x_data(1,i,:),N,1),'o-','Color',color_map(i,:));
    hold on;
end
for i=1:3
    plot(temp_valuex,-
180/pi*reshape(x_data(2,i,:),N,1),'x-','Color',color_map(i,:));
    hold on;
end
legend([...
    "Roll"+num2str([1:1:3]');...

```

```

    "Pitch"+num2str([1:1:3]')...
    ]);
ylabel("Angle [\degree]");
title("Lower Joint");
xlabel("Repeat [")

set(figure_handle,'Visible','On');
end

```